Distribution Automation and Smart Grid (EC9130)

DC–AC Three-Phase Inverter using Model Predictive Control

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1. Introduction

The increasing demand for renewable energy sources such as wind power has created the need for efficient and reliable power conversion systems. In wind energy applications, electrical power is often generated in DC form and must be converted into AC for practical use. This conversion is achieved using voltage source inverters (VSIs), which play a critical role in ensuring power quality and system efficiency.

Conventional inverter control techniques, particularly Sinusoidal Pulse Width Modulation (SPWM), are widely used due to their simplicity. However, SPWM-based systems suffer from inherent limitations, including high switching harmonics, dependence on passive filtering, and limited dynamic performance. These drawbacks result in increased Total Harmonic Distortion (THD) and reduced overall system efficiency.

In addition to control limitations, many traditional inverter designs lack integrated protection mechanisms. Over-voltage, over-current, and thermal conditions are often handled separately, leading to delayed response and reduced system reliability. This highlights the need for a control strategy that not only improves waveform quality but also enhances system safety.

Model Predictive Control (MPC) has emerged as a powerful alternative due to its ability to predict system behavior and directly control switching states. Unlike conventional methods, MPC eliminates the need for a modulation stage and enables real-time optimization of inverter switching. This results in improved waveform quality, reduced harmonic distortion, and faster dynamic response.

This project focuses on the design and simulation of a 1 kW three-phase inverter using MPC. The system includes an LC output filter and integrated protection mechanisms. In addition, a conventional SPWM-based inverter is implemented for comparison. The performance of both methods is evaluated using Total Harmonic Distortion (THD), demonstrating the advantages of MPC in high-performance inverter applications.

1.1 Research Gap / Problem Statement

The limitations of conventional inverter systems are:

- High harmonic distortion in output voltage
- Dependence on LC filters for harmonic reduction
- Indirect control of output voltage
- Limited dynamic response
- Lack of integrated protection mechanisms (over-voltage, over-current, thermal)
- Delayed fault detection and reduced system reliability

Therefore, an advanced control method is required to improve performance and ensure safe operation. Model Predictive Control (MPC) addresses these issues by providing real-time optimization and enabling integration of control and protection.

1.2 Objectives

The main objective of this project is to design and analyze a Model Predictive Controlled three-phase inverter for wind energy applications.

The specific objectives are:

- To develop a mathematical model of the inverter and LC filter
- To implement Model Predictive Control (MPC) for switching control
- To design and integrate protection mechanisms (over-voltage, over-current, thermal)
- To evaluate system performance using FFT and THD analysis
- To compare MPC performance with SPWM

1.3 Scope of the Study

This project focuses on simulation-based analysis using MATLAB/Simulink.

The scope includes:

- Modeling of a two-level three-phase inverter
- Design of LC output filter
- Implementation of MPC using MATLAB functions
- Development of protection system
- Performance comparison with SPWM

The study is limited to simulation; however, the proposed design is suitable for practical implementation.

2. System Overview

2.1 Overall System Description

The proposed system is a three-phase DC–AC inverter designed to convert a 400 V DC input into a balanced three-phase AC output. The system is developed in MATLAB/Simulink and incorporates advanced control and protection mechanisms.

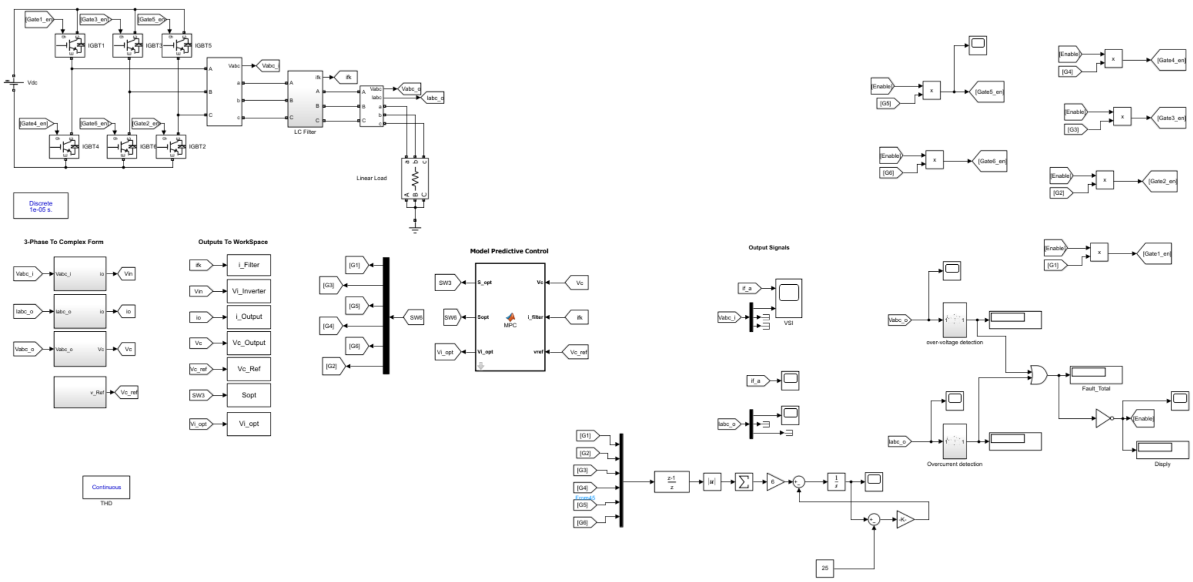
The main components of the system are:

- DC voltage source (400 V)
- Three-phase two-level inverter
- LC output filter

- Resistive load (50 Ω)
- Model Predictive Control (MPC)
- Protection system

The inverter generates a three-phase voltage which is filtered to produce a smooth sinusoidal waveform at the output.

2.2 Block Diagram Explanation



The system can be divided into three main parts:

Power Stage

- Includes DC source, inverter, LC filter, and load
- Responsible for power conversion

Control System (MPC)

- Receives feedback signals:
    - filter current if
    - capacitor (filter output) voltage Vc
- Determines optimal switching states

Protection System

- Monitors voltage, current, and temperature
- Generates fault signal
- Disables inverter when limits are exceeded

2.3 Power Stage Configuration

The inverter is implemented as a two-level three-phase bridge using six switching devices.

Each phase leg contains two switches operating in a complementary manner to prevent short-circuit conditions.

The DC link is formed using two 200 V sources connected in series: Vdc=400 V

This configuration provides a stable DC supply for inverter operation.

2.4 Signal Flow Description

The system operates in a closed-loop manner as follows:

1. A reference voltage is generated
2. Output voltage and current are measured
3. Signals are fed to the MPC controller
4. MPC predicts future output and selects optimal switching state
5. Gate signals are generated for the inverter
6. Output is filtered using LC filter
7. Protection system monitors system conditions and disables operation during faults

3. Design Specifications and Parameters

3.1 System Ratings

The inverter system is designed for approximately 1 kW power level, suitable for small-scale wind energy applications.

The key system specifications are:

- DC Link Voltage: 400 V
- Output Frequency: 50 Hz
- Reference Output Voltage: 188 V (phase peak)
- Rated Power: ≈ 1 kW

3.2 Voltage Selection

The reference voltage is derived from a standard three-phase system.

$$V_{phase,rms} = \frac{230}{\sqrt{3}} \approx 132.8\ V$$

$$V_{phase,peak} = 132.8 \times \sqrt{2} \approx 188\ V$$

Thus, the reference voltage is selected as:

$$V_{ref} = 188\ V$$

For a two-level inverter, the DC link must satisfy:

$$V_{dc} \geq 2 \times V_{phase,peak}$$

$$V_{dc} \geq 376\ V$$

A practical value of 400 V is selected to provide sufficient margin for stable operation and accurate control.

3.3 Load Selection

The load is modeled as a pure resistive load: R=50 Ω

The output current is:

$$I = \frac{V}{R} = \frac{188}{50} = 3.76\ A$$

This value is used in defining the over-current protection limit**.**

3.4 LC Filter Parameters

The LC filter is used to reduce switching harmonics.

- Inductance: L=2.4 mH
- Capacitance: C=40 µF

These values are selected to ensure effective filtering while maintaining system stability.

3.5 Control Parameters

The MPC controller operates in discrete time with a sampling period:

$T_s=10\ \mu s$

Justification

- Ensures accurate prediction of system behavior
- Captures switching dynamics
- Suitable for digital implementation

The reference voltage: $V_{ref}=188$ V is used as the target in the MPC cost function.

4. LC Filter Design and Analysis

4.1 Purpose of LC Filter

The output of the inverter contains high-frequency switching components due to rapid switching of power electronic devices. These components result in distortion and poor power quality.

An LC filter is used to:

- Reduce high-frequency harmonics
- Improve output waveform quality
- Provide a smooth sinusoidal voltage to the load

4.2 Inductor Equation

The inductor voltage is given by:

$$V_L = L\frac{di_f}{dt}$$

In the inverter system:

$$L\frac{di_f}{dt} = V_{inv} - V_c$$

where:

- V_{inv}: inverter output voltage

- Vc: capacitor (filtered) voltage
- if: filter current

The inductor limits rapid changes in current, reducing switching ripple.

4.3 Capacitor Equation

The capacitor current is:

$$i_c = C\frac{dV_c}{dt}$$

In the system:

$$C\frac{dV_c}{dt} = i_f - i_o$$

where:

- io: load current

The capacitor smooths the output voltage by absorbing high-frequency components.

4.4 Cutoff Frequency Calculation

The cutoff frequency of the LC filter is:

$$f_c = \frac{1}{2\pi\sqrt{LC}}$$

Substituting:

$$L = 2.4 \times 10^{-3}\ H, \quad C = 40 \times 10^{-6}\ F$$

$$f_c \approx 514\ Hz$$

Interpretation

- Fundamental frequency = 50 Hz
- Switching frequency ≈ 10 kHz

Thus:

$$50\ Hz < f_c \ll 10\ kHz$$

This ensures that:

- the fundamental component passes

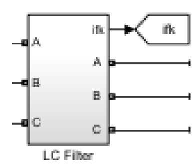
- high-frequency harmonics are filtered

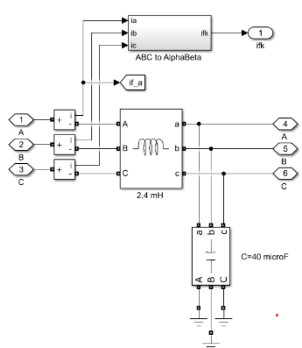
4.5 Design Justification

The selected LC filter values provide a balance between filtering performance and system stability:

- A lower cutoff frequency improves harmonic attenuation
- A higher cutoff frequency preserves the fundamental waveform
- The chosen values effectively reduce distortion without slowing system response

The filter parameters are also directly used in the system model, making them suitable for predictive control.





5. System Modeling

5.1 State-Space Representation

To implement Model Predictive Control, the inverter output stage is modeled using a state-space approach. The model is based on the LC filter dynamics.

The state variables are defined as:

The state vector is defined as:

$$x = \begin{bmatrix} i_f \\ V_c \end{bmatrix}$$

where:

- i_f = filter current
- V_c = capacitor voltage (filtered output voltage)

The control input is the inverter output voltage:

$$u = V_{inv}$$

The disturbance input is the load current:

$$d = i_o$$

Thus, the system can be written in the standard state-space form:

$$\dot{x} = Ax + Bu + B_d d$$

5.2 Continuous-Time Model

From the LC filter:

Inductor:

$$L\frac{di_f}{dt} = V_{inv} - V_c$$

$$\frac{di_f}{dt} = -\frac{1}{L}V_c + \frac{1}{L}V_{inv}$$

Capacitor:

$$C\frac{dV_c}{dt} = i_f - i_o$$

$$\frac{dV_c}{dt} = \frac{1}{C}i_f - \frac{1}{C}i_o$$

State-space form:

$$\dot{x} = \begin{bmatrix} 0 & -\frac{1}{L} \\ \frac{1}{C} & 0 \end{bmatrix} x + \begin{bmatrix} \frac{1}{L} \\ 0 \end{bmatrix} V_{inv} + \begin{bmatrix} 0 \\ -\frac{1}{C} \end{bmatrix} i_o$$

5.3 Interpretation of States

- if : represents current flowing through the filter inductor and determines system dynamics
- Vc : represents the filtered output voltage and is the main controlled variable

The controller aims to make Vc follow the reference voltage.

5.4 Physical Meaning

The model describes energy transfer in the system:

- The inverter provides input voltage Vinv
- The inductor controls current variation
- The capacitor smooths the output voltage
- The load consumes power

This model is used by the MPC to predict the future output voltage for different switching states.

6. Discretization of the Model

6.1 Need for Discretization

The system model derived in Section 5 is in continuous-time form. However, the MPC controller operates in a digital environment and updates at discrete time intervals.

Therefore, the model must be converted into a discrete-time form so that it can be used for prediction and control.

6.2 Discrete-Time Model

The continuous model:

$$\dot{x} = Ax + Bu + B_d d$$

is converted into:

$$x(k+1) = A_q x(k) + B_q u(k) + B_{dq} d(k)$$

where:

- Aq,Bq,Bdq are discrete matrices
- k represents the sampling instant

This equation is used by the MPC to predict the next state.

6.3 Implementation using cont2dis

The discretization is implemented using the following MATLAB function:

init_inverter.m | cont2dis.m

```
function [Aq, Bq, Bdq] = cont2dis(Ts,Lfilter,Cfilter)
    % Input filter model
    A = [0 -1/Lfilter;1/Cfilter 0];
    B = [1/Lfilter; 0];
    Bd = [0; -1/Cfilter];
    % Discretization of the input filter model
    [Aq,Bq]=c2d(A,B,Ts);
    [~,Bdq]=c2d(A,Bd,Ts);
end
```

This function converts the continuous system into discrete form using the sampling time: Ts=10 μs

6.4 Role in MPC

The discrete model is directly used in the MPC algorithm to:

- predict the future output voltage
- evaluate different switching states
- select the optimal control action

6.5 Sampling Time Selection

The sampling time is chosen as: Ts=10 μs

This ensures:

- accurate prediction
- proper capture of system dynamics
- efficient digital implementation

7. Model Predictive Control (MPC)

7.1 Introduction to MPC

Model Predictive Control (MPC) is an advanced control strategy that uses a mathematical model of the system to predict future behavior and select the optimal control action.Unlike conventional methods such as SPWM, MPC directly determines the inverter switching states without using a modulation stage. This enables improved waveform quality and reduced harmonic distortion.

7.2 Working Principle

At each sampling instant, the MPC controller:

- measures current system states (if, Vc)
- predicts future output voltage for all possible switching states
- evaluates a cost function
- selects the switching state that minimizes the error

This process is repeated at every sampling step.

7.3 Switching States

The three-phase inverter has 8 possible switching combinations. In the implemented controller, 7 active states are considered:

[100, 110, 010, 011, 001, 101, 111]

Each state represents the ON/OFF condition of the three inverter legs.

7.4 Voltage Vector Calculation

Each switching state produces a voltage vector given by:

$$V_i = V_{dc} \cdot \frac{2}{3}(s_a + a s_b + a^2 s_c)$$

where:

- $a = e^{j\frac{2\pi}{3}}$
- s_a, s_b, s_c are switching states

7.5 Prediction Model

Using the discrete model:

$$V_c(k+1) = A_q(2,1)i_f + A_q(2,2)V_c + B_q(2)V_i + B_{dq}(2)i_o$$

This equation predicts the next output voltage for each switching state.

7.6 Cost Function

The controller evaluates the error between reference and predicted voltage:

$$g = (Re(V_{ref}) - Re(V_c))^2 + (Im(V_{ref}) - Im(V_c))^2$$

The objective is to minimize this cost.

7.7 Optimization Process

For each sampling instant:

- all possible switching states are tested
- predicted voltage is calculated
- cost function is evaluated
- minimum cost is selected

7.8 Optimal Switching Selection

The switching state with the lowest cost value is selected as the optimal state.

This ensures that the output voltage closely follows the reference.

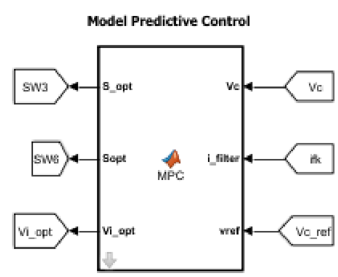
7.9 Gate Signal Generation

The selected switching state is converted into gate signals for the inverter switches.

Complementary signals are generated to ensure safe operation of upper and lower switches.

7.10 Advantages of MPC

- Direct control of output voltage
- Reduced harmonic distortion
- Fast dynamic response
- No need for PWM modulation
- Easy integration with protection systems



8. MPC Implementation (Code Explanation)

8.1 Overview

The Model Predictive Control (MPC) algorithm is implemented using a MATLAB Function block in Simulink. The controller uses the discrete-time model of the system to predict future output voltage and determine the optimal switching state at each sampling instant.

8.2 Initialization and Parameters

The controller receives the following inputs:

- Vc : capacitor voltage (output voltage)
- If : filter current
- Vref: reference voltage
- Vdc : DC link voltage
- Ts : sampling time
- L,C : filter parameters

The discrete system matrices are generated using:

```
function [Aq, Bq, Bdq] = cont2dis(Ts,Lfilter,Cfilter)
```

These matrices are used for prediction inside the controller.

8.3 Switching States Matrix

The possible switching combinations are defined as:

```
% To generate the switching states matrix
states = [1 0 0;
          1 1 0
          0 1 0
          0 1 1
          0 0 1
          1 0 1
          1 1 1];
a = exp(1j*(2*pi/3));
```

Each row represents a possible inverter switching state.

8.4 Voltage Vector Calculation

For each switching state, the corresponding voltage vector is calculated:

```
sw = (2/3)*(s(1)+a*s(2)+a^2*s(3));
Vi = Vdc*sw;
```

This converts switching states into a space vector representation.

8.5 Prediction Step

The next output voltage is predicted using:

```
% Output Voltage Prediction at instant k+1
Vc1 = Aq(2,1)*ifk + Aq(2,2)*Vck + Bq(2)*Vi + Bdq(2)*io;
```

This equation is derived from the discrete state-space model.

8.6 Cost Function Evaluation

The error between predicted voltage and reference is calculated as:

```
g = (real(vref)-real(Vc1))^2+(imag(vref)-imag(Vc1))^2;
```

This represents the tracking error.

8.7 Optimal State Selection

All switching states are evaluated, and the state with minimum cost is selected:

```
if (g < g_opt)
    x_opt = k;
    g_opt = g;
end
```

This ensures optimal voltage tracking.

8.8 Gate Signal Generation

The selected switching state is used to generate inverter gate signals:

```
S_opt = states(x_opt,:);
d_opt = not(S_opt);
Sopt = [S opt,d opt];
```

These signals control the upper and lower switches of the inverter.

9. Protection System

9.1 Overview

To ensure safe and reliable operation, a protection system is incorporated into the inverter. The system continuously monitors key electrical parameters and prevents operation under abnormal conditions.

The implemented protection system includes:

- Over-voltage protection
- Over-current protection
- Thermal monitoring

9.2 Over-Voltage Protection

The output voltage is continuously monitored and compared with a predefined threshold.

$$V_{threshold} = 206\ V$$

Justification

The nominal reference voltage is:

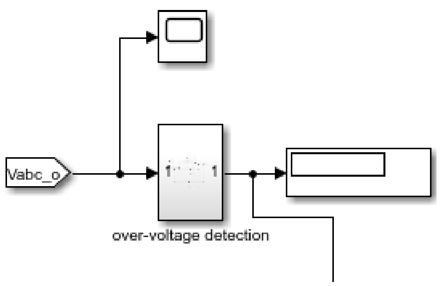
$$V_{ref} = 188\ V$$

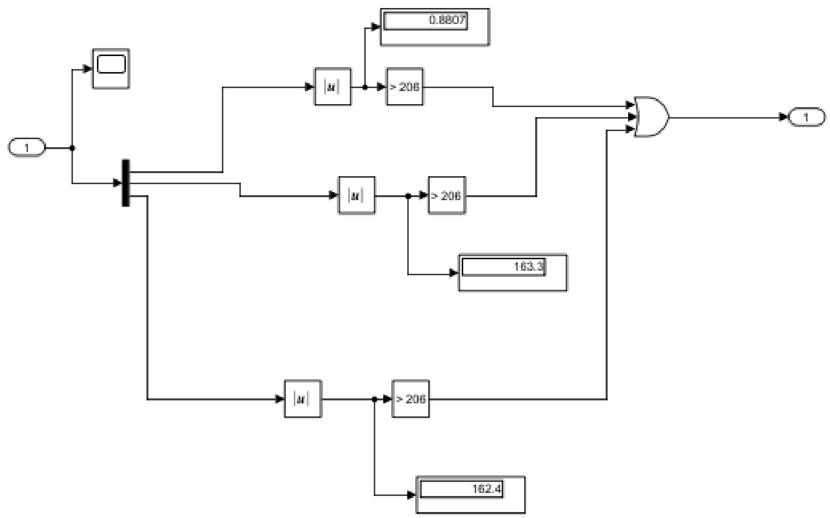
A safety margin of approximately 10% is added to account for transient conditions:

$$206\ V \approx 188 + 10\%$$

If the measured voltage exceeds this limit:

- a fault signal is generated
- the inverter is disabled





9.3 Over-Current Protection

The output current is monitored based on the load conditions.

The nominal current is:

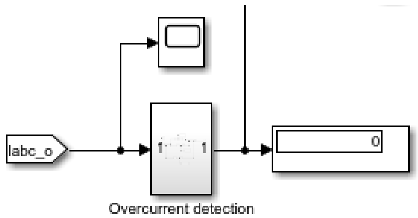
$$I = \frac{188}{50} = 3.76\ A$$

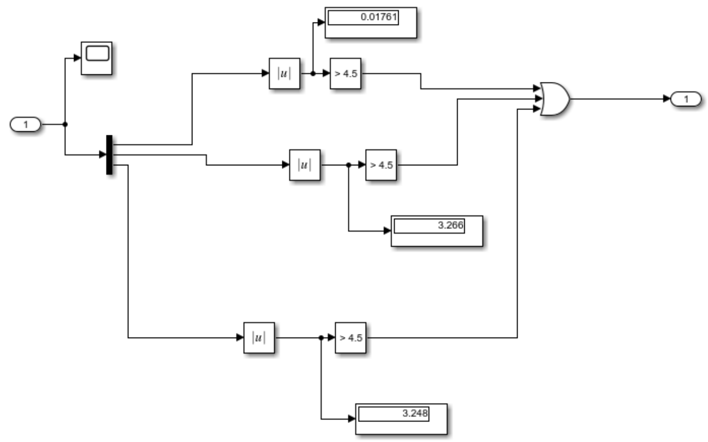
A safety margin is applied to define the threshold:

$$I_{threshold} = 4.5\ A$$

If the current exceeds this value:

- a fault condition is triggered
- the inverter switching signals are blocked





9.4 Thermal Monitoring

A thermal model is implemented to estimate the temperature of the inverter components during operation.

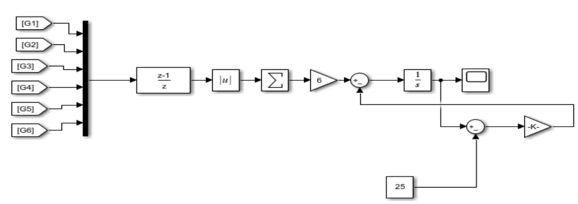
The temperature is described by:

$$C_{th}\frac{dT}{dt} = P_{loss} - \frac{T - T_{amb}}{R_{th}}$$

where:

- T: estimated temperature
- T amb: ambient temperature
- P loss: switching losses

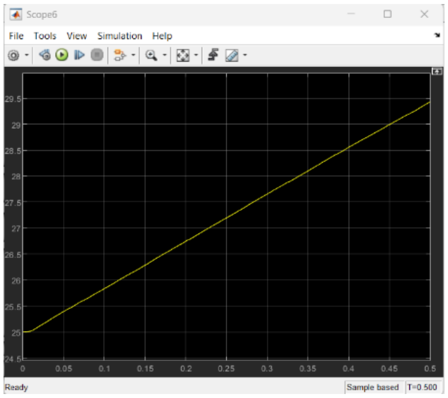
- Sum of gate signals → switching activity
- Gain (6) → loss scaling
- Integrator → temperature accumulation
- Feedback → cooling effect



A reference temperature limit is defined as:

$$T_{max} = 80^{\circ}C$$

In the current implementation, the thermal model is used for monitoring purposes only and is not directly connected to the inverter switching control.

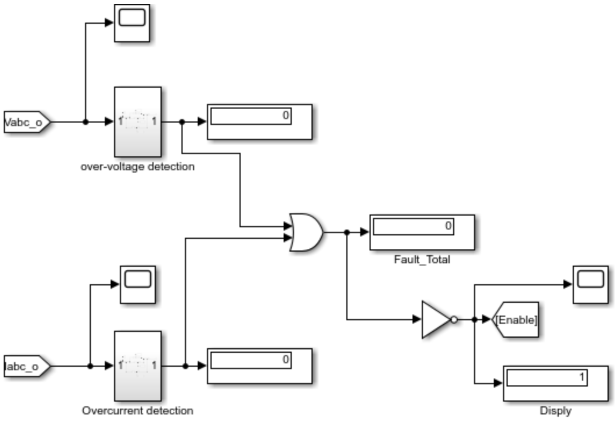


9.5 Fault Detection Logic

The electrical protection signals are combined using a logical OR operation:

$$Fault = V_{fault} \vee I_{fault}$$

This means that if either over-voltage or over-current condition occurs, a fault signal is generated.



9.6 Enable Signal and Gate Blocking

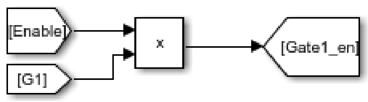
The fault signal is inverted to produce an enable signal:

$$Enable = \neg Fault$$

The enable signal controls the inverter operation:

- If Enable = 1, the inverter operates normally
- If Enable = 0, the gate signals are blocked

This ensures immediate shutdown during unsafe electrical conditions.



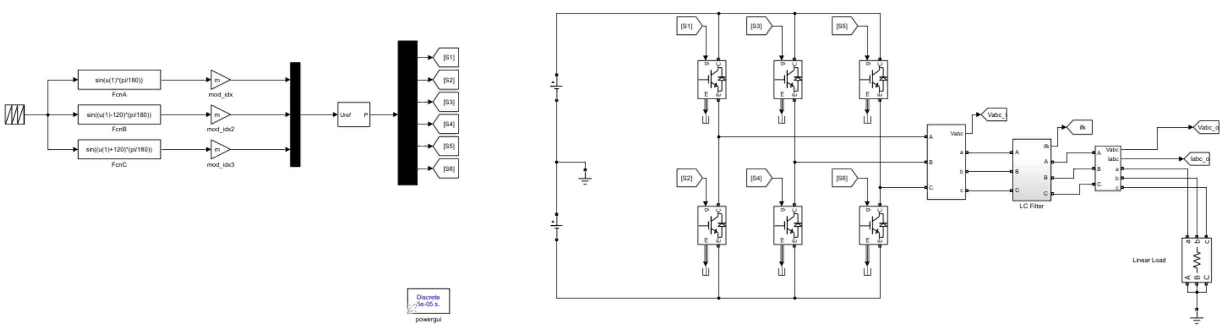
9.7 Remark on Thermal Integration

The thermal model is included for system monitoring and analysis. It can be integrated into the protection logic in future work to provide complete thermal shutdown capability.

10. SPWM Implementation

10.1 Overview

For comparison purposes, a conventional Sinusoidal Pulse Width Modulation (SPWM) based inverter is implemented. SPWM is a widely used control technique due to its simplicity and ease of implementation.



10.2 Working Principle

In SPWM, three sinusoidal reference signals (120° phase shifted) are compared with a high-frequency triangular carrier signal.

- If reference > carrier → switch ON
- If reference < carrier → switch OFF

This generates PWM pulses that control the inverter switches

10.3 Control Parameters

The SPWM system uses the following parameters:

- Modulation Index: m=0.95
- Switching Frequency: $f_{sw} = 10\ \text{kHz}$
- Fundamental Frequency: $f = 50\ \text{Hz}$

These values determine the output voltage amplitude and switching behavior.

10.4 System Structure

The SPWM system consists of:

- Three sinusoidal reference signals
- Carrier (triangular) signal
- Comparator block
- Gate signal generator

The generated pulses are applied to the inverter switches.

10.5 Limitation of SPWM

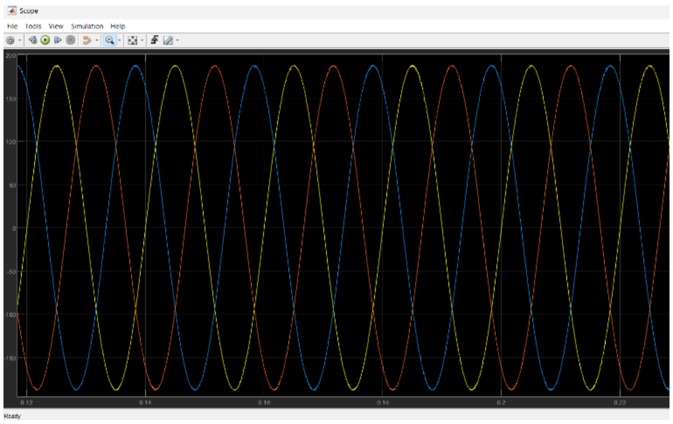
Although SPWM is simple, it has several limitations:

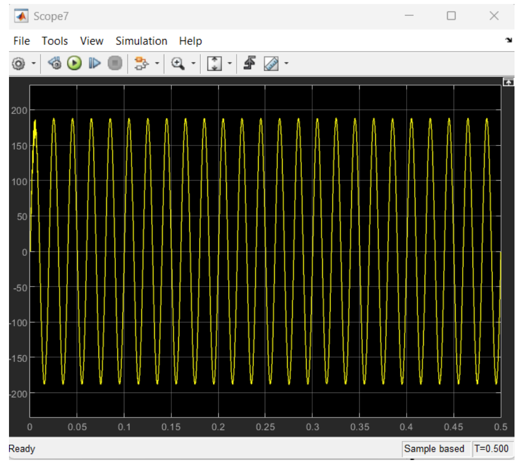
- Higher harmonic distortion
- Dependence on LC filters
- Slower dynamic response
- Indirect control of output voltage

These limitations make SPWM less effective compared to MPC for high-performance applications.

11. Results and THD Analysis

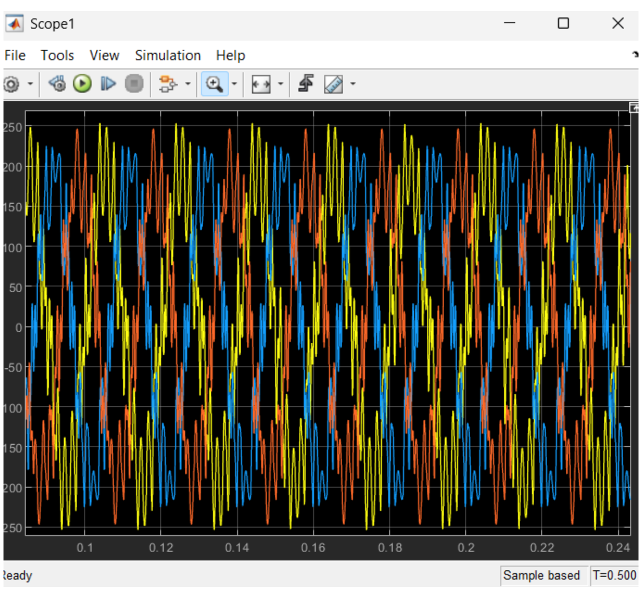
11.1 Output Waveform – MPC

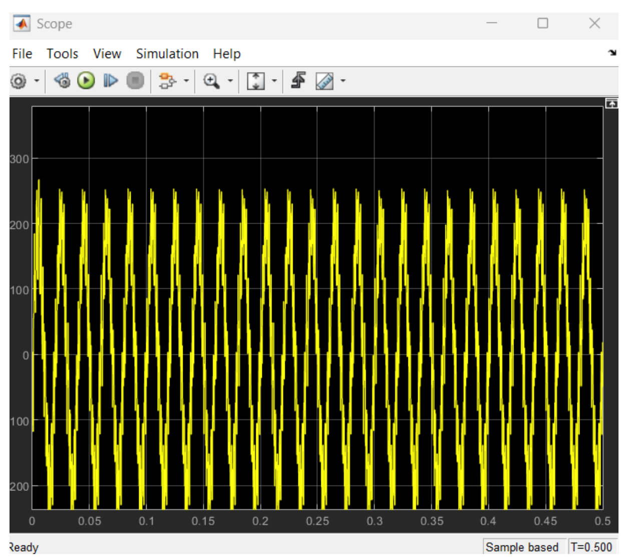




The output voltage obtained using Model Predictive Control is smooth and closely follows the reference sinusoidal waveform. The LC filter effectively removes switching harmonics, resulting in a high-quality output.

11.2 Output Waveform – SPWM





The SPWM output shows noticeable distortion due to switching harmonics. Although filtering is applied, the waveform is less smooth compared to MPC.

11.3 FFT Analysis Method

The Total Harmonic Distortion (THD) is evaluated using FFT analysis.

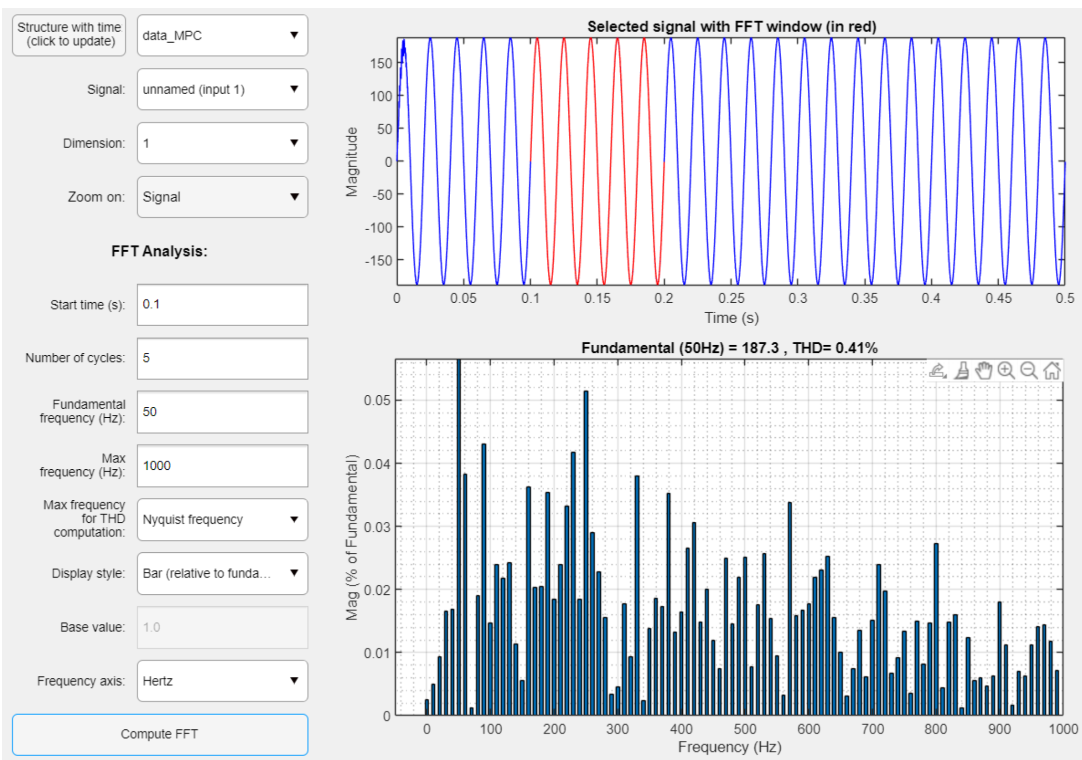
The analysis is performed with:

- Fundamental frequency: 50 Hz
- Number of cycles: 5
- Start time: 0.1 s

The FFT is applied to one phase of the output voltage.

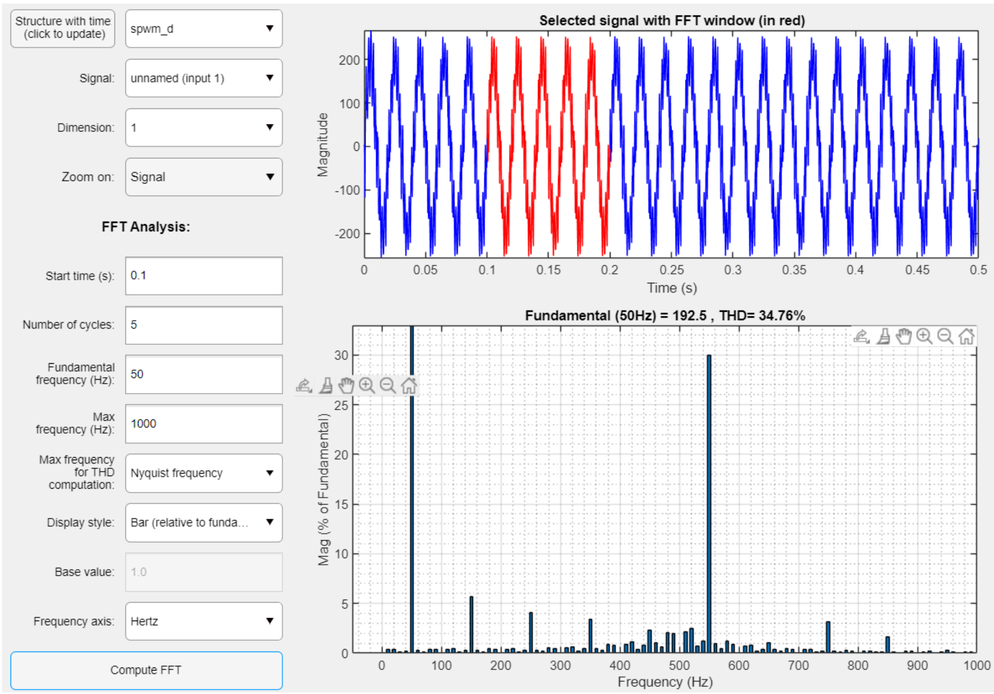
11.4 THD Results

MPC Result



- Fundamental Voltage: ≈ 187.3 V
- THD: ≈ 0.41%

SPWM Result



- Fundamental Voltage: ≈ 192.5 V
- THD: ≈ 34.76%

11.5 Comparison

Parameter	MPC	SPWM
THD	~0.41%	~30%
Waveform Quality	Very smooth	Distorted
Harmonics	Very low	High
Control Type	Predictive	Modulation

The MPC-controlled inverter produces significantly lower harmonic distortion compared to SPWM. The predictive nature of MPC allows direct optimization of switching states, resulting in superior waveform quality.

12. Conclusion

12.1 Summary of Work

In this project, a three-phase inverter system was designed and simulated using Model Predictive Control (MPC). The system includes an LC output filter and a protection mechanism to ensure safe operation. A conventional SPWM-based inverter was also implemented for comparison.

The mathematical model of the LC filter was developed and discretized for use in the MPC algorithm. The controller was implemented using a MATLAB Function block, where optimal switching states were selected based on prediction and cost minimization.

12.2 Key Findings

The results demonstrate that:

- MPC provides significantly lower harmonic distortion compared to SPWM
- The output voltage using MPC closely follows the reference waveform
- The LC filter effectively removes switching harmonics
- The protection system successfully prevents unsafe operating conditions
- Thermal behavior can be monitored for future integration

12.3 Final Conclusion

The MPC-based inverter shows superior performance in terms of waveform quality and control efficiency. The predictive nature of the controller enables real-time optimization, resulting in low THD and improved system response.

Compared to conventional SPWM, the MPC approach offers a more advanced and reliable solution for high-performance inverter applications.

12.4 Future Work

The system can be further improved by:

- Integrating thermal protection into the fault logic
- Implementing the system in real hardware
- Extending the controller for nonlinear or variable loads
- Optimizing computational efficiency for real-time applications

13. References

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